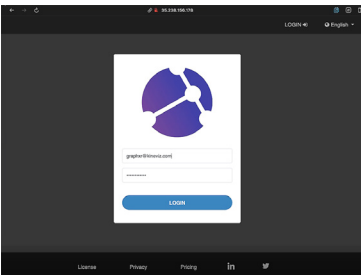




Kineviz Explorer for Spanner is a workspace for property graph visualization and analytics. Connect Spanner Graph databases to an Explorer project, run no-code queries, and explore the graph without moving or replicating data.

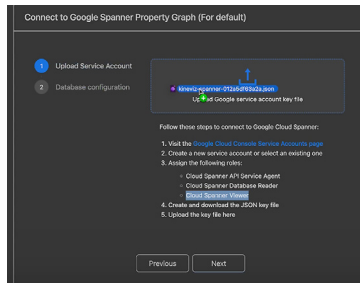
1. Launch from GCP Marketplace

1. Open the [Kineviz Explorer](#) product page
2. Click **Launch** to configure a deployment.
3. Enter a Deployment name.
4. Create a Service Account: under **Deployment Service Account**, click **New Account**. Enter
 - Service account name, ID and description.
 - Zone (us-central-c)
 - Under **Firewall**, click **Allow HTTP** and **Allow HTTPS** checkboxes.
5. Click **Deploy** and wait for deployment to complete.
6. Open the deployment **Details** tab and locate **Outputs**
 - Copy the **Admin Password** (click the Copy Link icon)
 - Click **Site URL** to open Kineviz Explorer login.
7. Log In to Kineviz Explorer
 - Enter your user ID and Admin Password, and click **Login**.
 - A Kineviz Explorer project is created, but is not yet connected with a specific database.



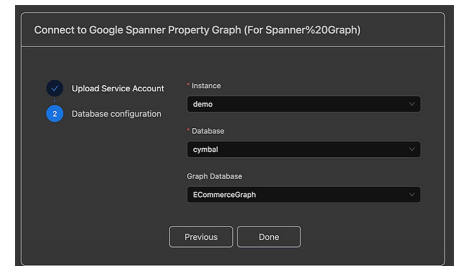
2. Obtain and upload a service account key

- The connection wizard links to your Google Cloud account, and lists steps to connect to a Spanner Graph database.
1. Click the link to visit the Google Cloud Console **Service Accounts** page.
 2. Open the service account created for the project (or create a new one).
 3. Under **Permissions**, assign 3 roles:
 - Cloud Spanner API Service Agent
 - Cloud Spanner Database Reader
 - Cloud Spanner Viewerand click **Done** to return to the Service Accounts page.
 4. Create and download the JSON key file
 - Click the 3 dots at the far right on the service account list item, and choose **Manage Keys**.
 - Click **Add Key**, and choose **Create New Key**, select **JSON** and click **Create**.
 5. Drag the downloaded key file to the connection wizard to upload, and click **Next**.



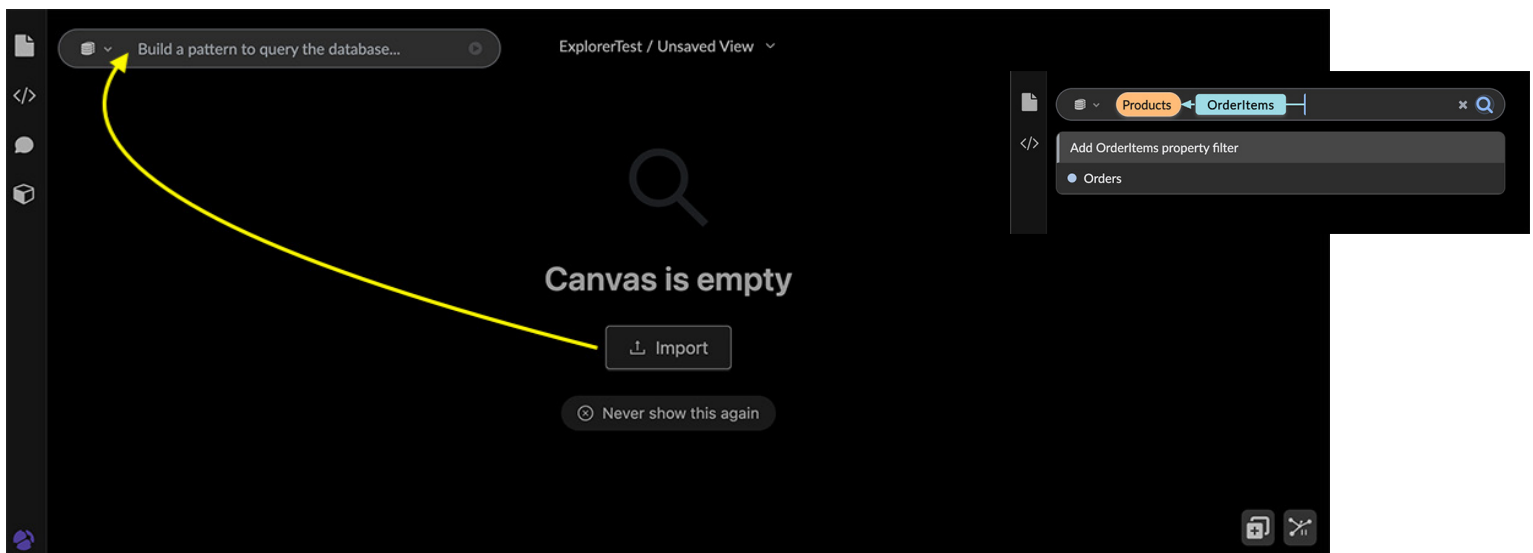
3. Review database configuration

- Database details are pre-filled.
1. Review to make sure the project will be connected to the expected database.
 - **Instance**,
 - **Database** name and
 - **Graph Database**
 2. Click **Done**.



The Kineviz Explorer project is now connected to the database.

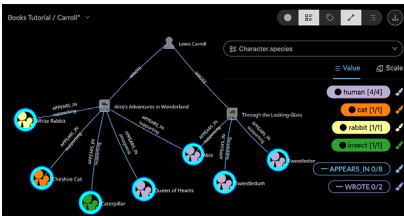
- The project canvas is empty when it first opens.
- To import data from the connected database, click in the Query bar to use Kineviz's point-and-click query builder.



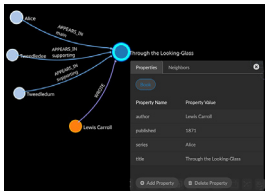


The Property Graph

- **Nodes** labeled by **Category**
- **Edges** labeled by **Relationship** and source and destination nodes.
- **Properties** of Categories or Relationships as **Property Name - Property Value** key-value pairs. To work with properties:
 - In the Legend, click **Property** for selectable lists of nodes or edges with unique property values.



- Double-click a node or edge for editable property **Information**.



- Open a **Table** of nodes (or edges) for a category (or relationship).
- Set property values as captions on nodes and edges.

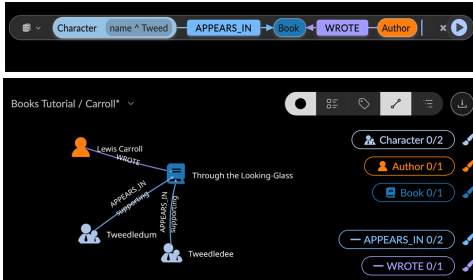
Query the Graph Database

Build and Run a no-code Query

1. Click in the Query/Search bar & choose Categories, connected Relationships and (optional) property value filters.

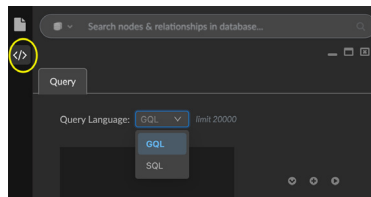


2. Click the query icon to run the query.



Enter and run a GQL or SQL query

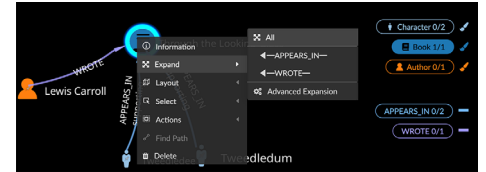
1. Open the **Query** panel.
2. Select **GQL** or **SQL**
3. Enter a query, and click the **Run** arrow.



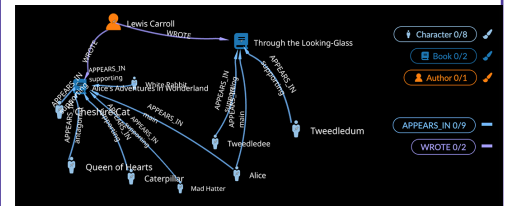
Expand with Relationships

- The **Expand** function queries for nodes in the database connected through one or more relationships to nodes already on the project canvas.

1. Select one or more nodes on the canvas.
2. Right-click and choose **Expand**.

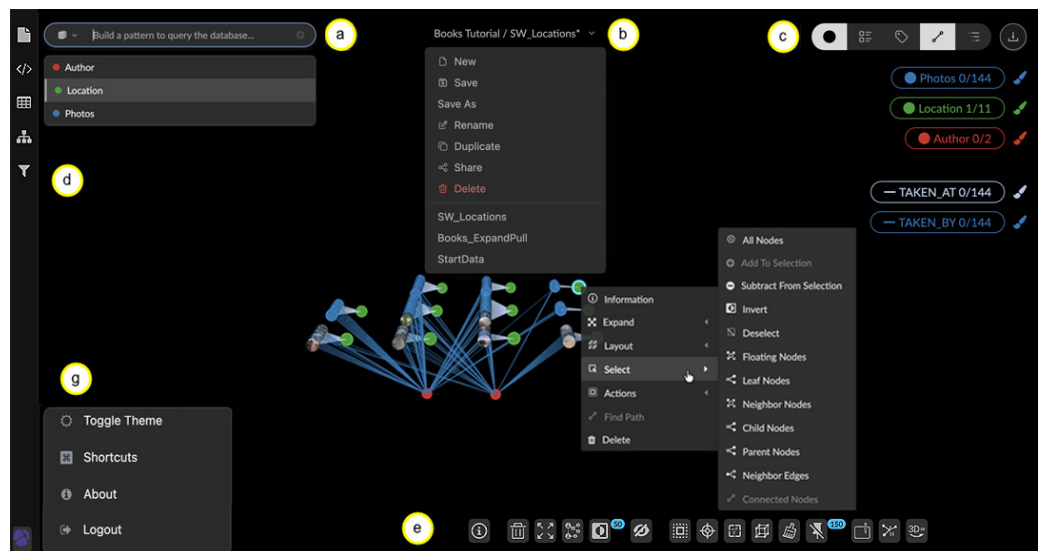


3. Choose
 - a single relationship to expand on
 - **All**, or
 - **Advanced Expansion**, for more control



The Project Workspace

- Query Bar**
- View Menu**
Save, Load, Rename, Duplicate, Share, and Delete Views
- Legend**
Select and Style by Category, Property, and Relationship
- Panel Menu**
Project, Query, Table, Layout, Filter
- Contextual Toolbar** icons
Toolbar only: Select Visible, Hide, Show Hidden, Center To, Fly Out, Quick Info, 2D/3D toggle.
- Right-click menu**
Expand, Layout, Select, Actions, Find Path, Delete
- Kineviz Logo Menu**
Toggle Theme, Shortcuts, About, Logout

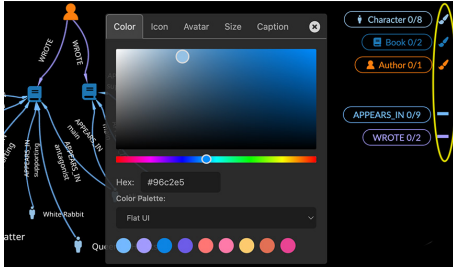




Style Nodes and Edges

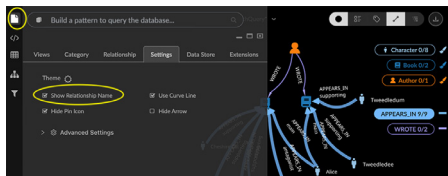
• Legend - Style Setting Dialog

1. Click the icon next to a category, relationship, or property.
2. Set **Color, Icon, Avatar, Size, Captions**



• Project>Settings

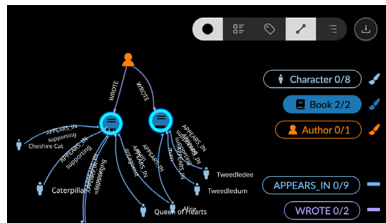
- Show Relationship Name checkbox



- **Advanced Settings** dropdown
- **Node** or **Edge** size scale
- **Caption** position

Navigate and Select

- **Pan, Zoom, Rotate, Select, Undo/Redo**, etc. Select **Shortcut** on the Kineviz logo menu for shortcut key reference.
- **Legend Selection** by category, property value or relationship. Click an item in the Legend list to select the nodes (or edges).



- Right-click **Select** and **Action** menus
 - > **Select**
 - **All Nodes, Floating** nodes, or **Leaf** nodes
 - **Invert** or **Deselect** current selection
 - **Neighbor, Child, Parent** nodes, or **Neighbor** edges of selected nodes
 - **Connected** nodes of selected edges
 - > **Action**
 - **Pin** or **Release** pinned nodes

• Toolbar-only actions

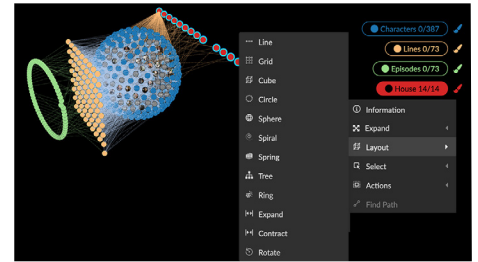
2D/3D toggle, **Quick Info** toggle, **Clear Graph**, **CenterTo**, **FlyOut**, **Select Visible**, **Hide**, **Show Hidden**, **Trace Neighbor**, **Delete Selection**.



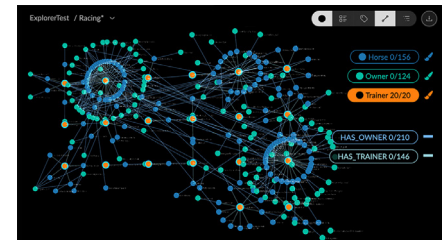
Lay Out the Graph

• Geometric layout

1. Select nodes to lay out
2. **Right-click>Layout**, and choose
 - **Line, Grid, Cube, Circle, Spiral, Sphere**, or **Spring** layout.



- Hierarchical **Ring** or **Tree** layout, centered on the nodes you selected.

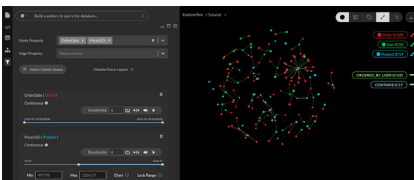


- **Expand** or **Contract** node spacing
- **Rotate** a selection counterclockwise 90 degrees

Filter and Focus

• Filter on Property Values

Open the **Filter** panel and choose node or edge property values to filter.



• Isolate subgraphs

Right-click>**Select>Neighbor, Child, Parent** nodes, **Neighbor edges** of selected nodes, and/or **Connected nodes** of selected edges

• Find and Isolate a Path

1. Select two nodes (click, then ctrl-click)
2. Right-click>**Find Path**



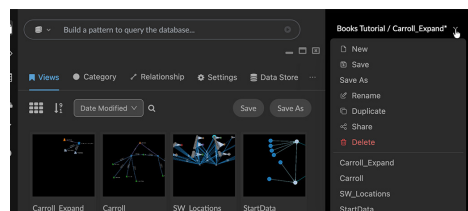
3. Click toolbar **Invert** and **Hide Selection** icons to hide data not on the path.

Save Graph Views

• Save a View from the view menu

- **Save**, or **Save As** saves the current view.
- **New** clears the graph & creates a new empty view.

• Load a saved view from the view menu



• Manage Saved Views

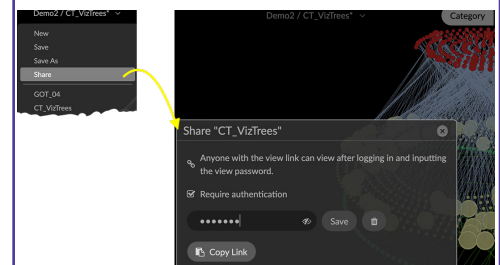
Open the **Project** panel and **Views** tab.

- Display as tiles or as a list.
- Sort by name, date created or date modified.
- Search for a view. Click the search icon to filter on view name.
- Share, Delete, or show view Information.
- Edit a view name.
- Click a tile or list to load a view to the canvas.

Share a View

• Share the view on the project canvas

1. Choose **Share** on the view menu.
2. Optionally, require authentication and enter a password.
3. Click **Copy Link**, and send it and the password (if any) to your colleagues.



• Share a saved view

1. Open the **Project** panel and **Views** tab, locate the view, and click its share icon.
2. Complete steps 2 and 3 above.



Shortcut Keys for Navigation, Selection, and Editing

Action	Shortcut	Action	Shortcut
Delete selection	delete	Select all nodes	ctrl+A
Clear graph	ctrl+shift+C	Deselect all nodes	alt+A OR double click empty space
Undo	ctrl+Z	Select a node or edge	left mouse click
Redo	ctrl+Y	Select multiple nodes and edges	shift+left mouse click-drag
History	ctrl + shift + H	Add a node to a selection	ctrl+left mouse click
Save	ctrl + S	Add multiple nodes to a selection	ctrl+left mouse click-drag
		Subtract a node from a selection	alt+left mouse click
Pan	left mouse click-drag OR 1+Left mouse click-drag	Subtract multiple nodes from a selection	alt+left mouse click-drag
Zoom	mouse wheel OR 2+left mouse click-drag	Select all orphaned (floating) nodes	shift+D
Fast zoom	ctrl+mouse wheel	Invert selection	I
Fly Out (Reset view)	alt+shift+R	Invert selection for selected node categories	shift + I
Center on a selected node	c+left mouse double click	Toggle Leaf Trimming	ctrl+L
Rotate (Manual)	right mouse click-drag OR 3+left mouse click-drag		
Toggle Rotate (Automatic)	ctrl+O	Full Screen	ctrl+F
Toggle Orbit (Automatic) around x- axis	ctrl+alt+shift+o	Hide menu	esc
Toggle Orbit (Automatic) around y- axis	ctrl+o		
Toggle Orbit (Automatic) around z- axis	ctrl+shift+o		
Stepwise Move	Arrow key click (right, left, up, or down)		
Move a selection	right mouse click-drag		
Toggle Pin / Release	ctrl+P		
Information	ctrl+I OR left mouse double click		
Expand with Relationships	ctrl+X		
Display Enhanced Table	shift + T		
Run Degree Centrality Algorithm	shift + ctrl + D		
Toggle Encrypt/Decrypt Text Property Values	shift + E		